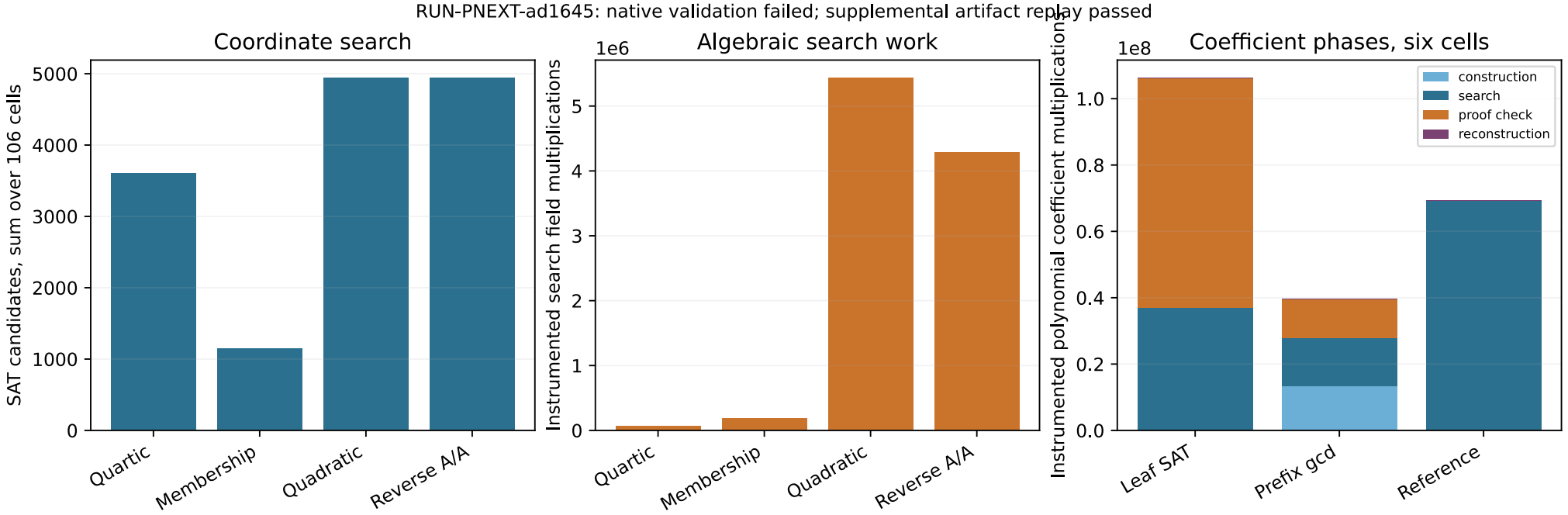
Certified SAT follow-up: retained observations

Observations only. The single native trial completed data collection but its frozen checker failed on an explicit-zero versus absent census counter. The original invalid_output receipt is preserved. A separate post-data checker changed only sparse counter equality and passed all retained model, certificate, witness and declared chart checks. This is exploratory replay, not a successful native run, independent scientific review, or hypothesis promotion. EXP-PNEXT-624f41 / RUN-PNEXT-ad1645. Native flow: one trial attempted, producer return 0, checker return 1. Ten aggregate export rows remain failed/excluded with verification_pass=0. Supplemental replay: 424 encoding arms (406 complete searches, 18 censored), 24 coefficient arms, 14,324 encoding certificates, 29,478 coefficient prefix certificates and 500,848 leaf certificates passed. The nine original synthetic controls and five sparse-counter integrity controls passed. Raw S3 objects were verified by fresh HEAD/GET/hash; native snapshot archival and scientific review remain pending.



Encoding inputs: E/F_17:y^2=x^3+2x+2 (order 19), E/F_97:y^2=x^3+x+4 (order 89); two- and three-bit x domains; all affine target x representatives with least nonzero y. Each arm has 106 cells and the same complete projected census, 85 coordinate pairs in aggregate. Search is separately bounded at degree 2, 256 rows, 128 SAT candidates, 512 PC calls. Quadratic input degree does not establish lower degree of regularity. These previously seen finite fixtures provide no held-out or cryptographic-scale inference.

Recorded counts and provenance

| Encoding | Complete/censored | SAT candidates | Search field multiplications | Online certificate multiplications | Final certificate multiplications | Full-census assignments |

| quartic | 106/0 | 3,599 | 69,422 | 6,630 | 6,630 | 4,240 |

| membership | 106/0 | 1,149 | 189,611 | 13,814 | 13,814 | 4,240 |

| quadratic | 97/9 | 4,944 | 5,436,704 | 362,242 | 362,242 | 473,600 |

| quadratic_aa | 97/9 | 4,944 | 4,286,333 | 362,242 | 362,242 | 473,600 |

| Coefficient arm | Full assignments checked | Prefixes | Construction mult. | Search mult. | Proof-check mult. | Reconstruction mult. |

| leaf_sat | 501,126 | 29,478 | 0 | 37,035,034 | 69,234,376 | 48,386 |

| prefix_univariate_gcd | 278 | 29,478 | 13,449,386 | 14,377,830 | 11,856,592 | 48,386 |

| exhaustive_reference | 501,126 | 0 | 0 | 69,279,657 | 0 | 48,386 |

| aa_reverse_prefix | 278 | 29,478 | 13,449,386 | 14,377,830 | 11,856,592 | 48,386 |

The prefix method retains 501,126 univariate root tests despite only 278 full coefficient checks. Relation coverage is five distinct affine x coordinates, with 155 generic and 123 fallback relations; repeated coordinates are outside scope.

Every coefficient arm additionally performs 2,240 fallback signed-point enumerations and 11,200 Jacobian additions. Construction, proof checking, recovery, exhaustive reference and supplemental verification are charged separately in the full report. Counters are instrumented subsets, not total runtime.

Quadratic A/A search multiplications differ with generator order; coefficient A/A field counts match but SAT decisions differ. No regularity, speedup or scalar-recovery inference.

Protocol a0a7009ad69a99ada238a5918d303ba8271e58f873d6986649691eae0c8d8cca; native authority 49ccff63d2039a708c276fccb3e8cd421c397d1b

Sources: arxiv.org/html/1503.08001; arxiv.org/html/2402.17927v1. All raw links, exact commands, controls, host facts and primitive counters are in report.md, report-data.json and export.json beside this PDF.